

DecodeLabs – Data Analysis Project 1 Report

Project Title

Advanced Exploratory Data Analysis (EDA) & Feature Engineering

Objective

The purpose of this project was to analyze and preprocess a raw dataset before machine learning. The workflow included exploring the dataset, handling missing values, identifying and removing outliers, and creating useful features to improve data quality.

Dataset Overview

Dataset Name: Dataset for Data Analytics.csv

Total Columns: 14

Initial Rows: 1200

Final Rows After Cleaning: 1192

Tools & Libraries

Python, Pandas, NumPy, Matplotlib

Step 1: Dataset Loading and Exploration

The dataset was imported using the Pandas library. Initial exploration was carried out with functions such as **head()**, **shape**, **info()**, and **describe()** to understand the dataset structure, data types, and statistical summary.

Step 2: Missing Value Assessment

The dataset was examined for missing values using the **isnull().sum()** function. Missing entries were identified and appropriate preprocessing techniques were applied to improve dataset quality.